Recent developments in microwave processing of minerals

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The demand for minerals and metals is increasing. However, it is well established that the impact of mining and subsequent processing operations must be reduced to meet future sustainability requirements. One way in which mining operations may reduce their impact is through the use of emerging science and technology. One such area which has shown particular promise over a period of several years is the use of microwave heating technologies to improve the efficiency of various mineral processing unit operations including: leaching, refractory gold ore treatment, grindability and liberation and coal grinding. Each of the above areas is reviewed and the main reasons that have either ensured or hindered development to a pilot or industrial scale are discussed. Opportunities for the future development of microwave technologies for the minerals industry are discussed and the review concludes by identifying a series of research challenges that must be met before scale up to an industrial level can be considered.

Keywords: Microwave, Energy efficiency, Comminution, Mineral processing

Introduction

The mineral resource and environmental industries are major employers and represent a significant proportion of GDP in the UK and throughout Europe. World consumption of minerals is also increasing, a trend likely to continue if the substantial and rising expectations of countries in the Pacific Rim, Central and South America, and Africa are to be met. The sustainable development and use of raw resources require new, better and more efficient processes. The recent Technology Foresight document from the UK Government's Natural Resources and Environment Panel[1] stated that new developments were required to improve and adapt existing unit processes. In particular, improvements in rock breaking and mineral liberation techniques and reductions in energy requirements for metal production processes were required.

In the United States, the situation is similar to the UK and Europe. The US Department of Energy has investigated through its Industrial Technologies Programme, the Mining Industry of the Future.[42] A series of roadmaps were drawn up to identify critical pathways to energy efficiency and sustainability within the mining sector. Of particular interest for this study is the roadmap for the development of mineral processing. This document projected that improvements in mineral processing technology would lead to the following benefits:

- Energy Efficiency – Improved energy efficiency per unit of output
- Environment – Reduced emissions per unit of output
- Health and Safety – Improved worker safety in processing activities
- Productivity – Reduced cost and higher value per unit of output
- Reserve Base – Increased reserves through technological advances and improved economics.

It is therefore clear that research over the next 20 years must address these issues. One such technology which address the needs of each of the points above is the use of microwave technology. The application of microwave energy for improving the energy efficiency of industrial processes is not new and several reviews in this area have appeared over the last few years.[2,17] Both of these papers concluded that although the use of microwave technology was attractive, it would only be economic under certain conditions whereby the key benefits of microwave heating over conventional heating methods were utilised. For example, faster heating rates, no direct contact between the heating source and the heated material, a high degree of selective heating, greater control of drying processes, reduced equipment size and the opportunity for the production of less waste.[17]

The use of microwave heating for processing minerals is also not a new concept and two reviews were published on the subject several years ago.[14,19] These papers offered a state of the art review (at the time) of applications of microwave heating in the minerals industry, including initial research and mineral heating studies, applications of microwave energy in thermally assisted comminution, studies in hydro and pyrometallurgy and a brief review of microwave treatment of coal.

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In addition, the paper by Haque[14] specifically reviewed the basic concepts of microwave heating, basic heating system design, temperature measurement and control. For a basic introduction to the area of microwave processing of minerals, it is to these papers that the reader is referred.

Kingman and Rowson[19] concluded that although the quantity of research carried out in the field of microwave processing of minerals was limited, the results published up to that point were encouraging with some potential for industrial implementation offered. However, before industrial implementation could be considered, further directed research was required in four specific areas, namely:

- Minimisation of energy input into the ore or mineral
- Frequency optimisation for maximum heating efficiency
- Data concerning the effects of sample temperature on dielectric properties
- Scale up to tonne per hour systems.

In his review, Haque[14] also concluded that microwave energy had the potential for application in mineral treatment and metal recovery processes. It was suggested that owing to the large scale of many mineral treatment processes (>30 000 t day^{-1}), a large number of parallel generators would be required to process the total plant feed. Therefore it was unlikely that the use of microwave energy would be economic unless high value products or low tonnage applications could be found, for example treatment of refractory gold ores, regeneration of carbon from carbon in leach operations and other roasting and smelting operations. Haque also concluded,[14] as did Kingman and Rowson,[19] that certain research and development was required before industrial implementation of the technology could be considered. The research and development requirements were listed as:

- Leaching at ambient temperature and pressure
- Pretreatment of refractory gold ore or concentrate under dry conditions as well as wet
- Dehydration of chemical products, e.g. $MgCl_2.6H_2O$
- Stripping of gold loaded carbon
- Decontamination of contaminated soil
- Decomposition of smelter gas
- Vitrification of inorganic waste.

It is interesting that two review papers in the area of microwave processing of minerals were published within several months of each other with each paper concluding that the future of the technology was optimistic. Both papers also concluded that specific research and development was required to aid in the industrialisation of the process. However, it is significant that each of the papers offered very different opinions as to where the optimal future applications of the technology lay, Kingman and Rowson[19] suggesting the use of microwave energy for selective heating to aid in mineral breakage processes and Haque[14] suggesting the use of microwave energy as a bulk heat source for say oxidation, dehydration or regeneration. An additional brief review of similar literature was presented in 1997.[38]

The present review will consider the evolution and innovations within the area of microwave processing of minerals since the publication of these previous reviews and compare the forecasts of these authors with the current advances and directions of the technology.

Applications of microwave energy for enhancement of mineral leaching processes

A detailed review of microwave assisted leaching was recently presented.[3] However, discussion of this type of process here is still valid as it is likely that as a result of environmental reasons, hydrometallurgical processes will attract significantly more interest over the coming years. Compared with pyrometallurgy, hydrometallurgical extraction of metals from their ores is potentially highly attractive. This attractiveness is attributed to economic, environmental and specific technical reasons. A major benefit of hydrometallurgical processes is that metals can be obtained directly in pure form from leach solutions. In addition, factors such as the relatively mild corrosion issues often suffered in hydrometallurgical plants compare favourably to refractory lining consumption in smelting. However, problems do arise during hydrometallurgical operations. These include: low recovery of the metal of interest, difficulty in solid–liquid separation and the influence of impurities on the ease of purification. Microwave assisted leaching has been investigated in an attempt to improve the yield of extracted metal. The unique characteristics of microwave heating processes listed above are certainly the drivers for potential implementation, i.e. reduced processing time, direct selective heating and a more controllable heating process.

The review by Al-Harahsheh and Kingman[3] considered the influence of microwave treatment on mineral leaching processes involving chalcopyrite, gold, coal desulphurisation, nickel, cobalt and manganese and the extraction of lead and zinc from their ores and compounds. The paper reported the results of over 70 papers; however, it was shown that there were no or limited industrial applications of the technology. This was suggested to be because of a lack of understanding of the influence of microwave energy on a particular leaching system. Several papers reviewed did however conclusively show that the rate or kinetics of the reaction were improved by the use of microwave heating compared to conventionally heated systems. The explanations which were presented in the paper to justify the acceleration of reaction rate under microwaves were:

- The existence of a non-thermal effect which reduces the activation energy of the reaction
- A superheating effect occurring during microwave heating making the temperature no longer representative of the reaction conditions
- A large temperature gradient between solids and liquids assisting the mass transport from the reaction interface as a result of the generation of large thermal currents
- An increase in surface area owing to cracks initiated in particles which contain minerals of different dielectric properties.

The major conclusion from the paper was that it was likely that the lack of reliable temperature measurement was the main reason for misunderstanding regarding any microwave effect in any given reaction system. Control and measurement of temperature is a major challenge in any microwave heated system because the electric fields present can cause serious errors in thermocouple measurements. In addition, it is also

nearly impossible to measure the actual temperature on specific mineral grains as inverse temperature gradients often occur on microwave heated particles (i.e. heat loss at the surface causes a higher core temperature, often leading to the statement that microwaves heat from the 'inside out').

Although the exact mechanisms by which microwave energy interacts with mineral leaching systems are not well understood, it is clear from the comprehensive review by Al-Harahsheh and Kingman[3] that the technology does have significant merit. There are however several areas which need to be addressed before the design of industrial reactors can take place, namely the relationship between frequency and penetration depth for highly ionic leach solutions and the electromagnetic design of reactors to ensure homogeneity of electric field.

Materials can be classified into three principal groups with respect to their interaction with a microwave field:

(i) Transparent or low loss materials where microwaves pass through without any losses
(ii) Conductors which reflect microwaves without any penetration
(iii) Absorbing or high loss materials, which absorb microwaves and dissipate it to heat depending on the value of the dielectric loss factor.

When a material contains two or more phases with different dielectric properties, a fourth category exists whereby microwave energy can selectively heat the adsorbent phase while passing through the transparent phase.[7]

Dielectric constant ε' and dielectric loss factor ε'' are used to express the dielectric response of materials in an applied microwave field. In simple terms, the dielectric constant measures the ability of a material to store microwave energy, i.e. the ability of material to be polarised. The latter measures the ability of material to dissipate the stored energy into heat.[10,34,40] They are expressed in terms of the complex dielectric constant ε^* as

$$\varepsilon^* = \varepsilon' - i\varepsilon'' \tag{1}$$

Penetration depth D_p (m) is the distance from the material surface where the absorbed electric field (e) falls to $1/e$ of the electric field at the surface. Penetration depth is inversely proportional to frequency whereas the greatest heating is achieved at high frequencies. For example, for water, the greatest rate of heating occurs at 18 GHz where the dielectric loss factor is at its maximum value. However, the penetration depth at such frequencies is so low that only the exterior of the material will heat.[35,40] D_p is given by

$$D_p = \frac{c}{2\pi f\sqrt{2\varepsilon'}\left[\sqrt{1+\tan^2\delta}-1\right]^{1/2}} \tag{2}$$

where f is frequency (Hz) and c is the speed of light (m s^{-1}).

Within the microwave frequency range, D_P varies from metres to millimetres depending on the applied frequency, temperature, chemical composition and microstructure.[4] Where the dielectric properties are high, such as in highly ionic salt solutions, the penetration depth can be of the order of a few millimetres at the standard microwave heating frequency of 2·45 GHz. This will obviously have serious implications for the design of any microwave heated industrial leaching system as penetration of energy even at lower frequencies such as 896 MHz or 433 MHz may only be of the order centimetres making the reactor small. The consequence of having a low penetration depth would be highly inhomogeneous heating across the reactor and a small zone which was actually heated by the microwave energy as opposed to heated by conduction. Therefore it is likely that the next stage of research required to advance the use of microwave energy in hydrometallurgical leaching systems will be a combination of reactor design and electromagnetic measurement and simulation to ensure that the full benefits of microwave heating as proposed in previous research (reviewed elsewhere[3]) can be realised.

Refractory gold ore treatment using microwave treatment

Gold is a noble metal and is therefore found in nature in its pure metallic form. Gold which has a grain size above which it is possible to recover with cyanide leaching alone is referred to as free milling gold (approximately >30 μm). Gold which has a grain size small enough to render it difficult to easily recover by alkaline cyanide leaching is referred to as refractory. The vast majority of refractory gold occurs within sulphide minerals such as FeS_2, FeAsS and $FeS_{(1-x)}$. Currently, most refractory gold ores are treated by roasting, pressure leaching or bacterial leaching to render the gold amenable to recovery by subsequent cyanide leaching.[12,13] Most iron rich sulphides are excellent microwave heaters with temperature increases of above 10 K s^{-1} even at relatively modest power levels such as 1–3 kW.[6,32,46] Laboratory scale tests to demonstrate the increase in gold recovery after cyanide treatment for different refractory ores have been carried out by various authors.[12,13,15,16] In each case, it was shown that microwave treatment before cyanidation increased the gold recovery significantly in many cases to up to 98% of the total. Kingman[20] carried out a detailed investigation into the influence of microwave pretreatment upon the mineralogy of a pyrite/marcasite based refractory gold ore from Lihir gold mine, Papua New Guinea. This work showed that significant oxidation of the sulphide phases occurred after treatment particularly at grain margins and there was clear evidence of the removal of sulphur from the system to cause the formation of $FeS_{(1-x)}$ and haematite as pyrite was oxidised. No attempt was made in any of these papers to extend the work from a laboratory scale and no real attempt was made to consider the economics of the process.

Scale up of the treatment of refractory gold ores using microwave heating has been extensively investigated in recent years with the development of a microwave heated fluidised bed system by EMR Microwave Technology Corporation in New Brunswick, Canada.[41] This work compared microwave heated fluidised beds to conventional refractory gold treatment processes and showed that the microwave process may have key advantages in terms of lower capital costs, reduced operating costs and the reduction or elimination of toxic byproducts requiring expensive treatment or remediation.

Enhanced separation as a result of microwave pretreatment

One highly promising area which has received a significant amount of attention is the enhancement of physical properties of a material as a result of microwave treatment.

Interestingly, according to the early reviews of Kingman and Rowson[19] and Haque,[14] the enhancement of magnetic properties of minerals was one of the first suggested applications for microwave energy in the minerals industry. Zavitsanos[49] studied microwave power adsorption at various frequencies for coal/pyrite mixtures. The work concluded that frequencies of 5 GHz gave the greatest differential heating between the coal and the pyrite. The separation of fine pyrite from coal is very difficult as pyrite generally has a very low magnetic susceptibility. However, if pyrite can be selectively heated within a transparent coal matrix, it is possible that it may be changed by oxidation to pyrrhotite ($Fe_{(1-x)}S$) which is significantly more magnetic. Early research in this area (up to 1997) was reviewed by Kingman and Rowson.[19] The major conclusion from this review was that microwave energy may offer an attractive method of converting pyrite to pyrrhotite for the purposes of desulphurisation. This conclusion agreed with the findings of a previous review. This study presented an overview of research carried out by the General Electric Company of North America which compared pilot scale desulphurisation studies with other methods of sulphur removal from coal. Microwave conveyor systems operating at 15 kW were used to selectively heat pyrite within run of mine coal. The microwave energy was found to selectively heat the pyrite in coal causing the formation of pyrrhotite. It was stated that pyrrhotite could be removed from the coal by low intensity magnetic separation. The quoted cost per tonne of coal treated was US$22, which compared most favourably with traditional technologies such as flue gas desulphurisation which had a stated treatment cost of US$26 per tonne of coal treated. Unfortunately, within this review, little technological details were provided to enable the issues of microwave energy consumption and scale up to be addressed. As coal is a bulk commodity, large tonnages will need to be treated (>50 t h^{-1}) for the process to be considered technically feasible. In fact, this conclusion was drawn by Kingman and Rowson in their review paper[19] when it was stated that although technically interesting, the issues of scale up need to be considered before the work could develop. It is somewhat surprising therefore that recent studies of coal desulphurisation through enhancement of the magnetic susceptibility of pyrite have appeared in the literature which do not address the area of scale up. In 2003, a study was carried out to investigate the influence of microwave treatment on the magnetic separation of pyrite.[44] Different size fractions of pyrite were heated in a domestic 850 W microwave oven operating at 2·45 GHz. The microwave treated samples of the -420 μm size fraction were subjected to magnetic separation at magnetic field intensities of 0·1, 0·3 and 0·5 T Unsurprisingly, it was found that pyrite was converted to pyrrhotite and γ-haematite and therefore magnetic separation behaviour was improved for treated material compared to untreated. It was stated in the paper that the maximum heating rates were obtained with the highest applied power levels and decreasing particle size. This is in agreement with a very similar study carried out by McGill *et al.*[32] The paper concludes that the use of microwave energy for the removal of pyritic sulphur from coal may soon be a commercial reality. Again, this is surprising given that the best experimental results within the paper were produced with a microwave energy input of over 3500 kW h t^{-1}! Microwave heating of coal for enhanced magnetic removal of pyrite was also recently investigated.[43] Turkish coal samples were subjected to magnetic separation at 2 T following treatment in an 850 W domestic microwave oven. It was found that increases in the magnetic properties of the iron sulphide phases were not sufficient to enhance the removal of a considerable amount of the pyritic sulphur. However, with the addition of 5% magnetite, an excellent microwave heater, microwave heating was enhanced and the pyritic sulphur content was reduced by 55·11% following magnetic separation at a field intensity of 2 T. In addition, the authors reported a 21·54% decrease in ash content coupled with a 20·9% increase in calorific value. No attempt was made to explain these observations and no comment was made on how the issues of scale up at economic microwave energy inputs would be addressed.

In addition to enhancement of the magnetic susceptibility of pyrite though microwave heating, significant studies have also been carried out on other mineral systems. The effect of microwave radiation on the magnetic properties of minerals was recently reported in detail.[22] In this work, a selection of common ore minerals (shown in Table 1) were subjected to microwave heating and the magnetic response of the mineral pre- and post-exposure determined by vibrating sample magnetometry. Changes in mineral structure as a result

Table 1 Influence of microwave exposure time and power level on change in magnetic susceptibility of common ore minerals (after Kingman and Rowson[22])

Mineral	Time of exposure, s	% Change in field strength
650 W		
Bornite	300	−82
Chalcopyrite	300	+103
Haematite	300	+5
Ilmenite	300	+5
Magnetite	300	−21
Pyrrhotite	300	+87
Wolframite	300	+10
1500 W		
Bornite	15	−95
Chalcopyrite	30	+85
Haematite	75	+14
Ilmenite	60	+17
Magnetite	15	−7
Pyrrhotite	90	−63
Wolframite	90	+25
3000 W		
Bornite	15	−94
Chalcopyrite	15	+49
Haematite	30	+31
Ilmenite	30	+35
Magnetite	15	−23
Pyrrhotite	15	−8
Wolframite	45	+44

of heating were determined by the use of X-ray diffraction analysis (XRD). The results of the magnetic survey are presented in Table 1.

As can be seen in Table 1, significant increases in magnetic susceptibility were apparent in chalcopyrite, ilmenite, haematite and wolframite, however, the magnetic susceptibility of bornite, pyrrhotite and magnetite were all reduced as a result of treatment. The study then showed that the two minerals which exhibited the largest changes in magnetic susceptibility, bornite and chalcopyrite, also underwent the largest changes in elemental iron composition during the treatment process. The paper concluded therefore that for significant change in magnetic susceptibility to occur, the chemistry of the mineral (in terms of the ratio of iron to other elements) must be altered during the heating process. It was also concluded that the treatment of minerals by microwave energy may mean that certain minerals not currently considered as targets for microwave treatment could become so.

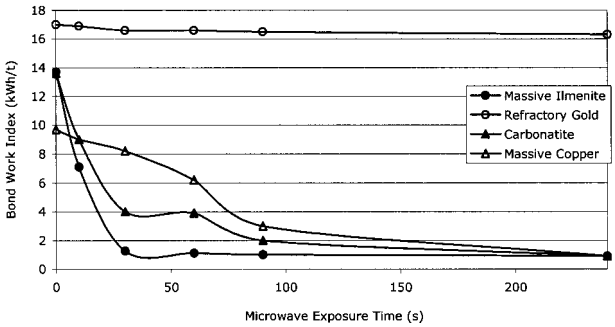
The above research by Kingman and Rowson[22] was a fundamental investigation into the influence of microwave treatment on the magnetic properties of pure mineral specimens. Several more applied studies have also been reported. The effects of microwave radiation upon the mineralogy and magnetic processing of a massive Norwegian ilmenite ore were presented in 1999.[21] Within this study, samples of ore were treated at 2·6 kW in a multimode microwave cavity for exposure times of 10, 30 and 60 s. Multimode cavities are the most commonly used form of microwave heating applicator. Mechanically, they are very simple, comprising of a closed metal box. The popularity of this type of cavity comes from not only the mechanical simplicity but also from their ability to process a wide range of workloads, in both size and electrical properties. To be considered as a true multimode cavity, the dimensions of the box must be several half wavelengths long in at least two orthogonal directions. When energy is launched into the cavity, a multiple mode pattern is set up, thus the name multimode cavity. A mode is a defined pattern of distribution of the electric and magnetic field components of an electromagnetic wave excited in a closed cavity where the electric and magnetic field components vary in a cyclical manner in the cavity space, so as to satisfy the boundary conditions at the cavity walls.[33] Higher electric field strengths can be generated in single mode cavities which comprise a metallic enclosure into which a microwave signal of the correct electromagnetic field polarisation will undergo multiple reflections. The superposition of the reflected and incident waves gives rise to a standing wave pattern that is very well defined in space. The precise knowledge of electromagnetic field configurations enables the dielectric material to be placed in the position of maximum electric field strength allowing maximum-heating rates to be achieved at all times. In the early evolution of microwave heating, such cavities saw little use. This was mainly because they lacked the versatility offered by multimode cavities. However, the development of electronic automatic tuning systems now means they are finding favour in industrial situations.

After treatment, the rock samples were ground to 100% passing 220 μm and treated by two-stage magnetic separation in a Boxmag Rapid high intensity wet magnetic separator. This was carried out to replicate the plant flowsheet as closely as possible. In the first stage of treatment, magnetite was removed from the ore with an applied magnetic field strength of 0·045 T and in the second, ilmenite was removed with an applied magnetic field strength of 1 T. After separation, the percentage of titanium in each sample was determined by colorimetry. It was shown that after 10 s exposure, a significant increase in both grade and recovery of titanium occurred compared to untreated material. This was suggested to be as a result of enhanced grain boundary fracture causing better liberation and also as a result of enhanced magnetic susceptibility of the titanium containing ilmenite phase. Further evidence for this was obtained by the fact that as exposure time was increased from 10 s, both grade of the concentrate and recovery of the titanium metal decreased as visible sample melting occurred reducing the magnetic susceptibility of the valuable ilmenite phase to the point where it was very difficult to recover. The paper concluded with a statement regarding the need to further reduce the amount of microwave energy used to achieve the benefits. It was suggested that this could be done by using higher powers for shorter durations or alternatively pulsed energy delivery.

The effect of heat treatment (both conventional and microwave) on chalcopyrite was investigated by Sahyoun *et al.*[39] Chalcopyrite is an industrially important mineral which forms an essential component of many copper ores. Currently, owing to its very low magnetic susceptibility, magnetic separation is not considered as a viable concentration route. It was shown within the paper that as a result of treatment in conventional furnaces for varying times, the proportion of material classed as magnetic after passing over a high intensity magnetic roll separator increased significantly. X-ray diffraction analysis of the treated samples showed major phase changes which were suggested to be responsible for the increase in magnetic susceptibility of the sample. Identical samples were also microwave treated in a multimode cavity and again the proportion of material recovered to the magnetic fraction was shown to increase when compared to untreated material. However, in the case of the microwave treated material, subsequent XRD analysis did not show any major phase changes. A full explanation for this was not presented.

Intensification of magnetic separation efficiency and also leaching of copper ores by microwave treatment were also the subject of another paper.[29] In this work, samples of Slovakian chalcopyrite and tetrahedrite concentrates (tetrahedrite is a multi component sulphide which contains copper, zinc, iron, mercury, nickel and cobalt) were heated in a domestic microwave oven at a power of 900 W. The magnetic susceptibility of non-treated and treated samples was determined, with variation in microwave exposure times of between 0 and 40 s. After 30 s exposure, the magnetic susceptibility of treated chalcopyrite was shown to increase by about 30%. Again, no mention was made in the paper of how the research could be utilised in an industrial situation.

In a very recent study, the modification of the magnetic properties of siderite ore by microwave energy has been investigated.[50] Samples of siderite (25·1%Fe and 9·6%SiO_2) were treated in a 900 W domestic microwave oven for varying time periods. After 10 min



1 Change in Bond work index versus microwave exposure time for given ore types (after Kingman *et al.*[24])

of heating, a change in magnetic susceptibility occurred which was found to become more pronounced after 15 min total exposure. Using XRD, the degree of magnetite (a ferromagnetic mineral) was shown to increase with increasing microwave exposure time to the point where after 30 min exposure, the sample contained magnetite as the major phase. Again, no conclusions were presented regarding the microwave energy requirement for the process and details of how scale up may be achieved.

Microwave assisted comminution

There is no doubt that it is in the area of microwave assisted comminution that most progress towards an industrial application of microwave energy in the minerals industry has been made since the publication of the reviews by Kingman and Rowson[19] and Haque.[14]

Before publication of the above reviews, many authors had presented work which showed that increased grindability of ores could be achieved by microwave pretreatment. However, while most of these were technically very interesting, they suggested that microwave assisted comminution was unlikely to be economically viable.

A detailed investigation of the influence of mineralogy on microwave assisted grinding was presented in 2000.[24] In this work, detailed mineralogical analysis of various commercially exploited ores was carried out and the textural relationships between the individual mineral phases established. Each ore type was then subjected to microwave energy for varying time periods and the change in relative work index with microwave exposure time quantified. It was concluded that the increase in grindability of the ores after treatment was directly related to the specific minerals present (heater or non-heater), grain size of the heated phase and also the degree of dissemination of the heated phase. A summary of the grinding results produced for the different ore types examined is shown in Fig. 1.

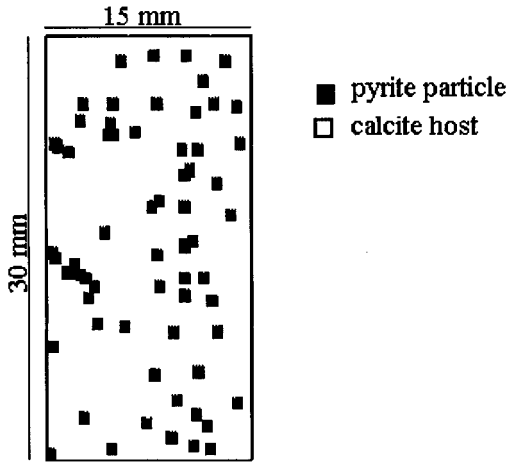
It was shown that significant improvements in grindability could be achieved at a microwave energy input of 14·3 kW h t^{-1}. This was compared to a grinding energy requirement of approximately 20 kW h t^{-1}. This is the first published account of microwave assisted grinding being carried out at even close to economic microwave energy inputs. Still, however, the question of scale up was not addressed. Also in 2000, further work was published by the same authors which considered the effect of microwave energy on the processing of Palabora copper ore.[23] In addition to



2 Microwave induced intergranular fracture (after Kingman *et al.*[23])

enhanced grindability, this paper also demonstrated that increases in recovery of copper after froth flotation could be achieved. The increases in recovery and grade were suggested to be a result of different heating in the sample between the different contained mineral phases. Scanning electron microscopy was used to determine whether any grain boundary fractures were present within the ore. It was found that significant intergranular fracture was present in the ore, even after 10 s treatment. Such fracture, shown between copper (chalcopyrite and bornite) and iron bearing (magnetite) phases is shown in Fig. 2. This was particularly true for short durations of heating (<10 s). Above this time, reductions in grade of concentrate and recovery of copper were found to occur which the authors suggested were as a result of surface oxidation owing to high temperatures.

Interestingly, this paper also attempted to show what would be the likely result of microwave treatment upon an operating grinding circuit. The open circuit rod mill, closed circuit ball mill grinding plant was modelled using a steady state flowsheet modelling package, USIMPAC. Simulations were carried out using operating plant data to calculate the circulating load within the ball mill circuit. The Bond work index of the ore was then reduced and the impact on circulating load calculated. It was shown that if the grindability could be reduced to 2 kW h t^{-1} (as was suggested by the testwork programme), then half of the current grinding circuit could be removed. This was a first attempt to try and determine the likely influence of microwave treatment on full plant performance.

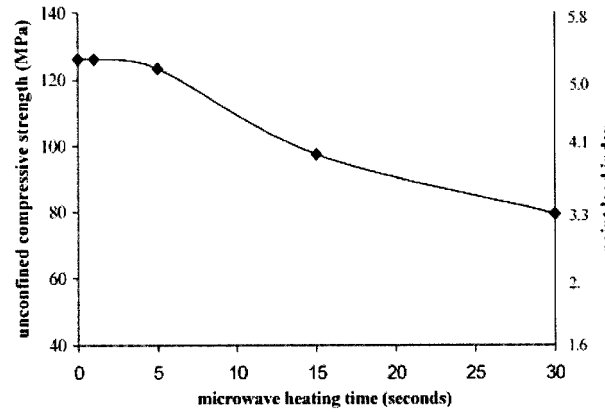


3 Model domain used to determine influence of applied power density on microwave assisted breakage (after Whittles *et al.*[47])

Altered surface chemistry as a result of microwave treatment was addressed by Vorster *et al.*[45] In this paper, the authors showed that significant changes in grindability could still be achieved without significantly altering the surface chemistry of the ore. A massive copper sulphide ore from Portugal was microwave treated for various times at a power level of 2·6 kW in a multimode microwave cavity. The grindability change in the material was calculated and was found to be 70% greater than an untreated sample with identical original size distribution. Comparative froth flotation studies were then carried out utilising treated and untreated ore. Despite using long exposure times (90 s), no change in the flotation performance of the treated material was made and the authors concluded that microwave treatment did not influence the surface chemistry of the ore. This is in disagreement with the findings of Kingman *et al.*[23] who reported significant reductions in copper recovery above and beyond 10 s microwave exposure. The paper concluded that pilot scale demonstration of benefits was required to derive a full techno-economic appraisal of the technology. A throughput of 500 kg h^{-1} was suggested to be ideal but no justification for this value was presented and no detail of the pilot scale test regime was given. The issues of passing large quantities of material continuously though a microwave cavity were also not addressed.

From the above papers and those reviewed in other papers, it is clear that microwave assisted comminution is technically possible, however the influences of important microwave variables such as power density or heating rate have not been quantified at this time.

The most significant breakthrough in the reduction of microwave energy input for microwave assisted comminution came by understanding the influence of power density on microwave assisted breakage.[47] In this study, an ore was simulated whereby the microwave absorbing phase was represented by pyrite, within a microwave transparent host represented by calcite. The influence of applied electric field strength (or power density) was assessed by numerical simulation. Simulations were undertaken using 2D finite difference modelling techniques of a theoretical 15×30 mm sample of calcite host rock containing 10% volume of discrete 1 mm^2 pyrite blocks. The model domain used is illustrated in Fig. 3.

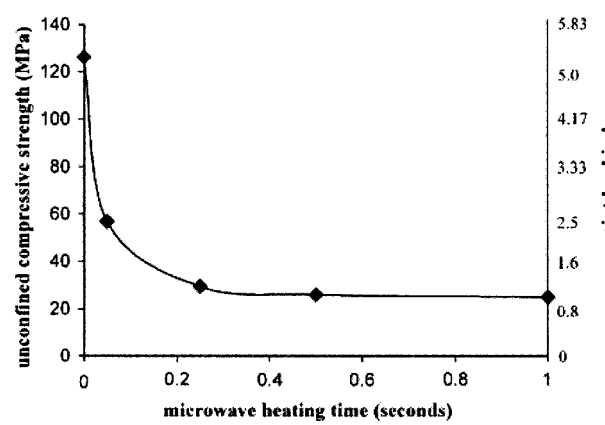


4 Simulated change in UCS versus microwave heating time for samples treated at 1×10^9 W/m^3 of heated phase (from Whittles *et al.*[47])

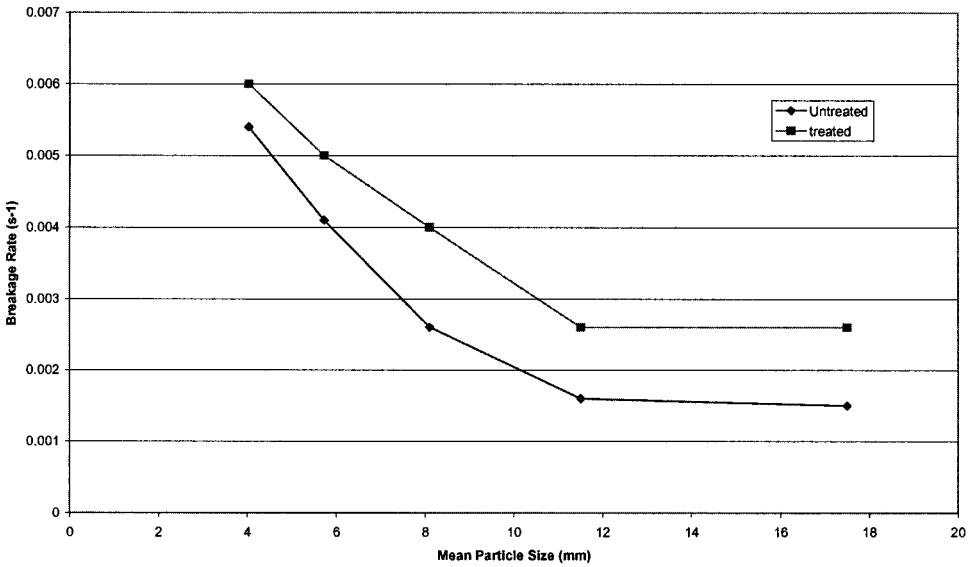
The simulations modelled the microwave heating, thermal conduction, thermal expansion, thermally induced fracturing and strain softening, and finally uniaxial compressive strength (UCS) to predict the effect of microwave heating on the strength of the ore material. Standard correlations were then used to relate the UCS to standard measures of crushability such as t_{10} (see Napier-Munn[36] and references therein for further information on such tests). Simulations were carried out at two power densities which were suggested to be typical of peak values which could be found in standard multimode and single mode cavities, namely: 1 GW m^{-3} for the multimode and 100 GW m^{-3} for the single mode. For the lower power density, simulations were run assuming microwave heating times of 1, 5, 15 and 30 s and then the resultant UCS values were plotted versus microwave exposure time. The resultant plot is reproduced in Fig. 4.

For the higher power density, simulations were run assuming heating times of 0·05, 0·25, 0·5 and 1 s. The resultant values of theoretical UCS for each sample were then plotted against the corresponding microwave exposure time. The resultant plot is reproduced in Fig. 5.

From a comparison of Figs. 4 and 5, the influence of power density can clearly be seen with significantly more weakening occurring in the samples exposed at 100 GW m^{-3} for 0·05 s than in the sample exposed for



5 Simulated change in UCS versus microwave heating time for samples treated at 1×10^{11} W/m^3 of heated phase (from Whittles *et al.*[47])



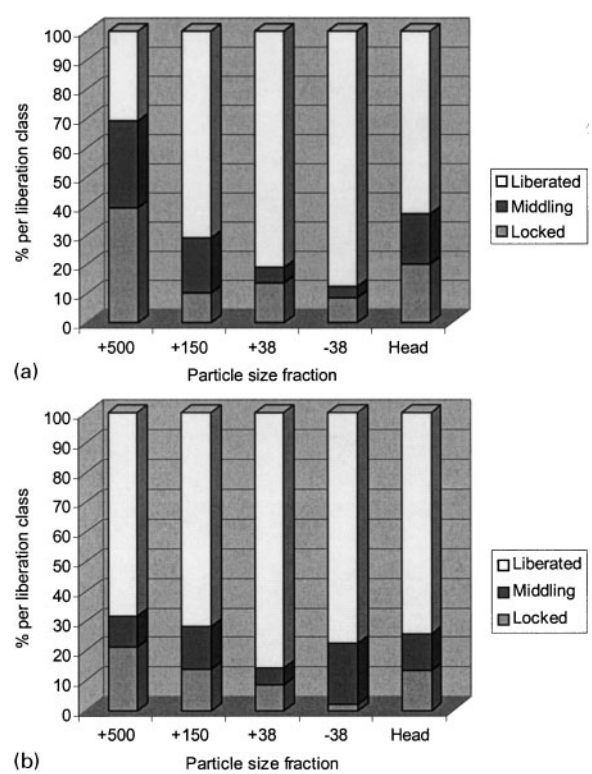
6 Plot of specific rate of breakage versus mean particle size for treated and untreated samples (after Kingman *et al.*[25])

30 s at 1 GW m^{-3}. Using thermal maps, it was shown that at the higher power density, heat transfer by conduction was minimised, with very little heat transfer away from the heated phase into the transparent phase. This gave much greater temperature gradients between the heated and non-heated phases than in the case where the sample was 'cooked' slowly for a long duration.

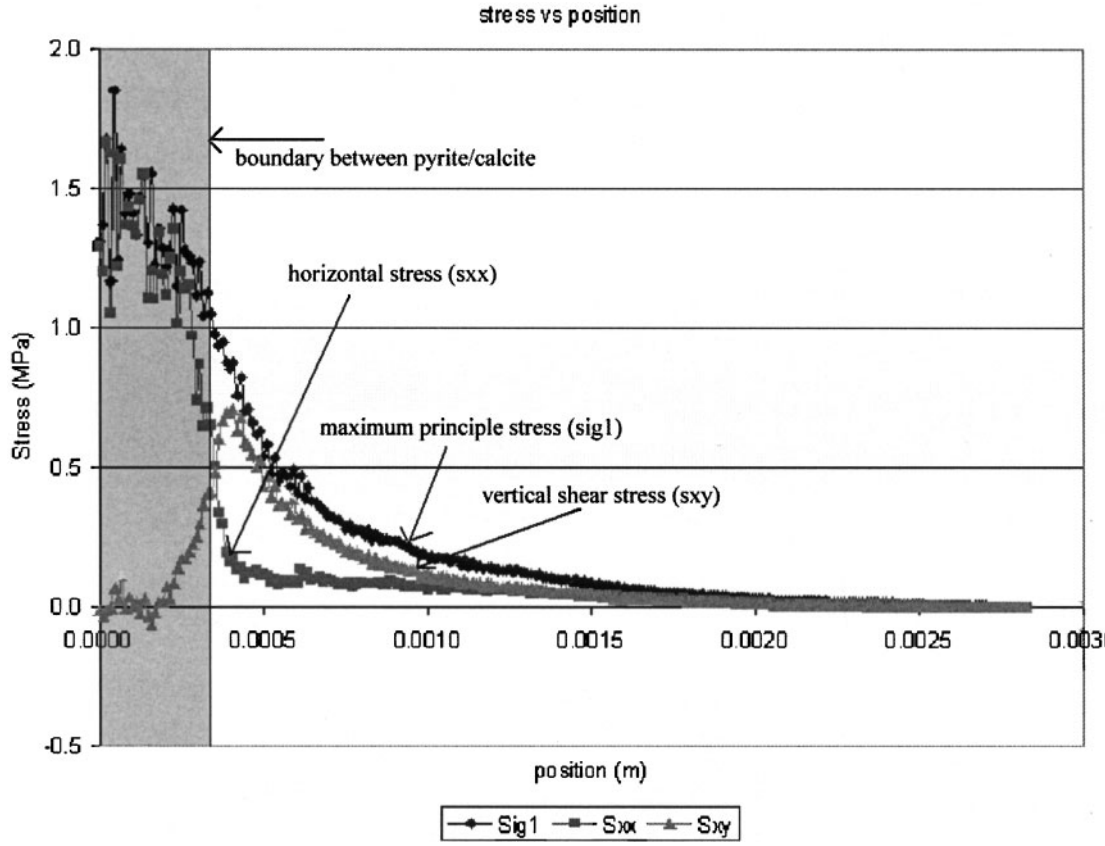
The paper concluded that by increasing the power density, significantly greater thermally induced stresses and strains were induced for a much lower microwave energy input. The paper also concluded that the use of high power density cavities (such as single mode) may make microwave assisted comminution economic for the first time. The first experimental validation of the work presented by Whittles *et al.*[47] was presented only very recently.[25] In this study, the influence of high electric field strength microwave energy on the processing of copper carbonatite ore was elucidated. It was shown that very short exposures (<0·1 s) at 15 kW in a single mode TE_{010} microwave cavity could lead to significant reductions in ore strength as determined by point load strength tests. Comparative drop weight tests were also carried out to determine any potential change in required comminution energy for microwave treated material. It was shown that reductions in required comminution energy of over 30% could be achieved for microwave energy inputs of much less than 1 kW h t^{-1}. In order to quantify the influence of microwave treatment on change in grindability, comparative specific rate of breakage studies showed that significant increases in grindability particularly for the larger size fractions (>12 mm) could be achieved, again for a microwave energy input of less than 1 kW h t^{-1}. Interestingly, it was also shown that there was a clear relationship between increase in grindability and particle size with very little change in particles of less than 3 mm. This was suggested to be as a result of the surface area of small heated particles becoming large as a function of their volume. This is shown in Fig. 6.

Enhancement in liberation of the valuable copper sulphide phase was made by QEM.SEM (quantitative analysis of materials with scanning electron microscopy) analysis.[37] Treated and untreated material was broken by impact with a breakage energy of 1 kW h t^{-1}. The resultant daughter particles were then screened at 500, 150 and 38 μm and the degree of liberated copper minerals in each size class determined (liberated was taken as greater than 80% copper mineral by area). The full liberation spectra for untreated and treated samples are shown in Fig. 7. When treated and untreated fractions were compared, it was shown that over 100% more copper sulphide was liberated in the treated +500 μm fraction than the untreated.

The paper concluded that, for the first time, significant changes in ore breakage behaviour and liberation characteristics had been demonstrated at



7 Liberation spectra for copper sulphide minerals for *a* untreated and *b* treated copper ore (after Kingman *et al.*[25])



8 Profile plot showing variation of stress across entire particle; shaded area shows regime within heated particle; vertical (Sxy) and maximum principal stresses (Sig1) both show compressive peaks towards centre of pyrite; shear stress (Sxy) shows peak value just outside grain boundary (after Jones *et al.*[18])

microwave power inputs which were potentially economic.

The work to some degree validated the theoretical studies of Whittles *et al.*[47] who suggested that by minimising residence time and maximising applied field strength or power density, thermal damage would be significantly increased.

Further evidence to support this theory was also recently presented.[26] In this study, samples of Swedish lead–zinc ore were microwave treated before strength testing and change in strength with exposure time determined. Comparisons in strength were then made between treated and untreated material. Using a multimode cavity, it was shown that strength could be significantly reduced in 0·5 s when using 15 kW of applied microwave power. Lower powers in this type of cavity were found to be less effective. Further preliminary tests were then carried out utilising a TE_{010} single mode microwave cavity. Samples of identical ore were exposed at 10 kW for 0·1 s and strength tested. Again significant strength reductions were noted, again indicating that high electric field strength is vital for the failure of ore.

While the above papers clearly demonstrate the importance of power density in microwave assisted comminution, they do not fully explain the microwave induced failure mechanism. A very recent study has again used numerical simulation to understand microwave assisted comminution.[18] A quasi static thermomechanical 2D model of a theoretical two-phase mineral (based again on calcite and pyrite) was constructed. The aim was to simulate the thermal stresses generated and subsequent thermal damage when a rock particle was exposed to high electric field strength microwave energy. As in the paper by Whittles *et al.*,[47] the modelling was used to simulate the extent of microwave heating, thermal conduction and resultant expansion, strain softening and thermal damage. The developed model system consisted of a 2 mm × 2 mm calcite host containing a variable size circular pyrite phase. For different assumed power densities within the pyrite phase, the stress regime was analysed and it was concluded that fractures were most likely to occur around grain boundaries between transparent and absorbent phases. This is illustrated in Fig. 8.

The major conclusions from the paper[18] were:

- The stress regime inside the heated phase was compressive while outside the boundary, shear and tensile stress predominated
- Material weakening is due to elevated shear stresses outside the grain boundary, exceeding the strength of the material
- Liberation can be enhanced, as a result of rocks being much stronger in compression and also the fact that the surrounding matrix holds the heated grain in confinement
- As the heated particle size decreases, more energy is required to sufficiently raise the temperature such that thermal stresses of a sufficient magnitude to damage the rock are generated.

The last point was particularly interesting as this agreed with the observations of Kingman *et al.*,[25] in that a reduction in observed 'effect' occurred with particle size for a fixed power density (or applied power level).

Microwave enhanced grinding of coal

The grinding of coal before combustion in a pulverised fuel (PF) burner requires a significant amount of power.[8] For example, the power required to grind a standard bituminous coal to PF in a Babcock Hitachi 10E10 coal pulveriser was found to be 266 kW for a feedrate of 8·4 kg h^{-1}.[9] This equates to approximately 6–8 MW of power to run the coal mills in a typical large power station. As shown above for metal ores, increasing the grindability of the coal is highly desirable and, while there are several different ways of achieving this end, some are more practical and economic than others. Preheating a coal in a standard oven has been shown to reduce grinding resistance[11] although the data presented showed that this was unlikely to be economic at large tonnage scales. As bulk heating the coal sample was not effective, it is not surprising that a significant body of work has been carried out utilising the selective heating benefits of microwaves for enhancing the grindability of coal.

The work carried out before 1998 has already been reviewed by Kingman and Rowson.[19] The major conclusions of that review were that, while coal grindability could certainly be enhanced by microwave treatment, the microwave energy inputs which were being used to achieve the reductions were uneconomic. In addition, as for the mineral ore grinding, few ideas were presented for the treatment of coal at large tonnage scale. Since the publication of that review, there have been some important breakthroughs in the enhancement of coal grindability using microwave energy and these will be reviewed here.

In 2000, an attempt was made to understand the effect of microwave radiation on coal grindability.[30] In the study, coals of varying rank were exposed to microwave energy to quantify the effect on grindability. Increases in grindability of up to 50% were reported after 5 min microwave exposure. These increases in grindability were suggested to be the result of two distinct fracture mechanisms, namely: the differential expansion of ash and pyrite within the transparent coal matrix and inherent moisture within the coal structure changing phase creating significant pressure. The authors also assessed the influence of microwave treatment on calorific value and reported little or no change. The final recommendation from the paper was that pilot plant scale testwork be carried out to allow an accurate techno-economic analysis of the process.

In 2001, an attempt was made to further understand the influence of microwave energy on coal.[31] Dielectric measurements of various UK coals showed that electrical permittivity decreases with coal rank (dry, mineral free basis), however, the moisture and mineral matter content increased the bulk coal electrical permittivity. The explanation for the increase in permittivity for low rank coals was that they contained more moisture, an excellent microwave absorber. Samples were tested over a wide range of frequencies but the authors claimed that the values of real and imaginary permittivity did not change significantly.

Much of the previous work, while quantifying the influence of microwave treatment, did not consider the effect of heating on the physical and petrographic characteristics of coals. A recent study has however addressed this.[27] In this study, a UK steam coal was sieved into several size fractions and then treated with high electric field strength microwave energy for a short (<10 s) duration, to investigate how the grinding resistance changed in comparison with untreated coals and also how the chemical petrographic compositions were altered after exposure. The paper showed that the short exposure to microwave energy gave a significant increase in grindability and, as for metal ores, these changes were again related to particle size. After detailed analysis of the coal structure, both pre- and post-treatment, the authors concluded that the effect upon the petrographic characteristics was negligible.

Increased coal grindability as a result of microwave pre-treatment at economic microwave energy inputs has also been presented very recently.[28] In this study, five coals from different parts of the world were ground and sieved into four size fractions (3·35–4·7, 6·7–9·5, 9·5–19 and 19–22·4 mm). The vitrinite reflectance of the coals ranged from 0·35 to 0·76% and the vitrinite content ranged from 55 to 96% (on a volume basis). Approximately 300 g of each size fraction was heated in a TE_{010} single mode cavity operating at 8 kW with an exposure time of 0·1 s, the settings being chosen to represent industrially relevant conditions, i.e. sufficient microwave power density and residence times to ensure high throughput. The paper demonstrated for the first time that significant increases in the grindability of coals could be achieved with microwave power inputs which were potentially economic in the context of electrical power generation. The final conclusion from the paper was that excellent results had been achieved with a non-optimised cavity design and that further reduction in the required microwave energy could be achieved with better designed systems.

Discussion

The present study has reviewed the developments in microwave processing of minerals and coal over the last 5–6 years. Before this time, a significant body of work had been carried out in this field although much of it was exploratory and lacked detailed understanding of the fundamental interaction of microwave energy with the material of interest. This in turn led to interesting technical results but, in almost every case, very poor economics when scale up was considered.

Since the late 1990s, the approach by several researchers has been different. An effort has been made to understand how microwave energy interacts with the material, how dielectric properties change with temperature and material content and how issues of scale up can be addressed. This greater understanding has led to significant process benefits being presented but crucially at potentially economic microwave energy inputs.

Industrial applications of microwave energy are still several years away and a considerable amount of research has still to be carried out to make the use of the technology a reality. This research falls into several key areas which are briefly discussed.

The optimisation of electric field pattern with a microwave cavity to ensure the correct degree of 'treatment' is highly important. Just as many domestic microwave ovens utilise turntables or mode stirrers to ensure that heating homogeneity is achieved in cooking, homogeneity is vital in industrial processing, be it rock

cracking or gold concentrate roasting. This to ensure that the desired 'effect' is achieved in the total sample and quality targets can be maintained. To produce heating homogeneity in mineral processing is a severe challenge as the bulk dielectric properties of the material will change many times a minute owing to the inherent variability of natural rock materials. Therefore a multi-disciplinary approach is recommended that links disciplines such as physics, electrical and electronic engineering, mineral processing, chemical engineering and microwave engineering. Such an approach has already been shown to provide step changes in understanding and therefore the development of proposed industrial systems.[5]

The second major challenge for the industrialisation of microwave technology in the mining industry is developing systems that are compatible with a harsh plant environment. The majority of current microwave installations are used in relatively benign environments such as food processing factories. A mining site is very different and issues such as dust, dirt and remote location must be considered by any microwave equipment manufacturer who wishes to meet the potential demand for microwave equipment in this industry.

One issue which was almost always not addressed in the papers reviewed here was the issue of compatibility of materials handling systems with microwave cavities at scale. Although the use of conveyor systems is well established with microwave systems (particularly multimode cavities)[32] for the mining industry, this is unlikely to be the case as a result of the issues of compatibility with high electric fields and highly abrasive materials. A recent paper has investigated the residence time distribution of material within a microwave cavity flowing under gravity.[48] It was shown that by feeding material vertically under gravity though a single mode cavity and controlling the discharge rate with a vibratory conveyor, control over flowrate could be achieved.

The major challenge for the industrial implementation of microwave technology in the mining industry is still without doubt the minimisation of required energy input and this is particularly true for microwave assisted comminution. The articles reviewed for the present study have shown that by minimisation of heat transfer from the heated phase to the bulk, the rate of stress rise within the ore lattice will be more significant. They have also shown that the best way to do this in practise is by using high electric field strength microwave energy applied for very short times. The ultimate way of applying short high power durations of microwave energy is through the use of pulses. The use of pulsed microwave energy is very common, albeit at different frequencies to that used for the work presented in the present study. Common applications include particle accelerators, radar systems and cancer therapy machines. By the storing and rapid discharge of energy, these systems can generate very high peak powers ($>$100 MW) with low mean powers ($<$100 kW). Necessarily, the discharge time is short (1–4 μs) and therefore potentially the throughput of ore could be massive (greater than several thousand tons per hour). This means that the overall microwave energy consumption could potentially be very low as total energy consumption is a function of microwave power, residence time and mass in the cavity. Pulsed treatment of ores should without doubt be the future research direction in this field.

Conclusions

The latest developments in the use of microwave energy in the mining industry have been reviewed. It has been shown that a knowledge of the fundamental interaction of microwave energy with mineral is vital for the development of systems which will operate economically. The major areas of research have also been reviewed and it has been shown that microwave assisted comminution and coal breakage will be major candidates for the first industrial installations. Although the future of microwave technology in these areas looks promising, a significant amount of further research is required before industrial application can occur.

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